

Motivation

- Growing Use of VPAs (Voice Personal Assistants) Across Demographics:** VPAs are used globally by diverse populations, yet the technology struggles to respond equally well across races, genders, and age groups.

- Biases in ASR Technology:** Speech recognition systems often exhibit bias, influenced by training data and algorithms that prioritize certain demographic groups, leading to disparities.

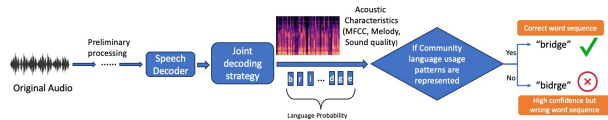


- Limitations of Current Testing Methods:** Traditional testing is labor-intensive and limited in scope, making large-scale and diverse demographic testing challenging.

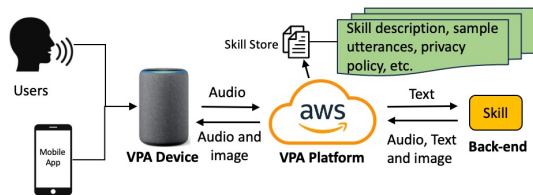
- Need for an Automated Solution:** Developing an automated, scalable tool like AcousticScope addresses these limitations, enabling comprehensive and efficient bias testing across demographic groups.

Biases Exist in ASR System

- Lack of Diversity in Training Data:** ASR models are often trained on datasets that predominantly feature speech from specific demographic groups (e.g., well-educated, middle-aged White speakers). This imbalance limits the system's ability to recognize speech patterns used by other racial and ethnic groups effectively.
- Imbalanced Data Distribution:** ASR models, often built using deep neural networks, optimize performance based on the most represented data in the training set. This means that groups with fewer samples may not receive adequate feature training, resulting in lower recognition accuracy for these groups.

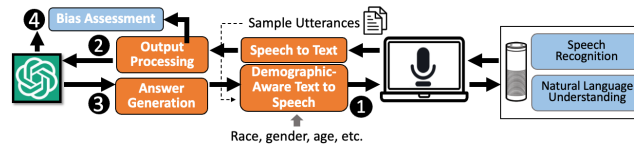


Voice Interaction in VPAs



Working principle and skill response process of VPAs devices

System Overview



1. Demographic-Aware Speech Processing: The process begins with demographic-aware Text-to-Speech components that consider factors such as race, gender, and age, aiming to generate and interpret speech samples inclusively. We use Lovo AI to generate a variety of different voices.

2. Speech Output Processing: Sample utterances from users and ASR system are processed through Speech-to-Text technology, converting spoken language into text that can be analyzed by ChatGPT.

3. Answer Generation: After speech is converted into text, a response is generated by ChatGPT which formulates an answer based on the input.

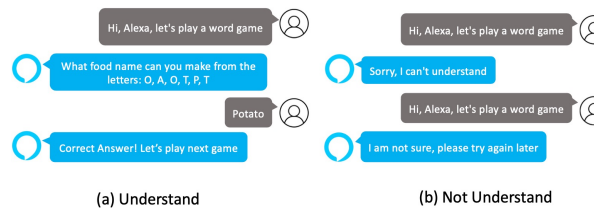
4. Bias Assessment: The generated output is processed, and bias assessment is conducted to evaluate potential biases in the responses related to demographic variables.

Biases Assessment

1. Data Flow of Bias Assessment



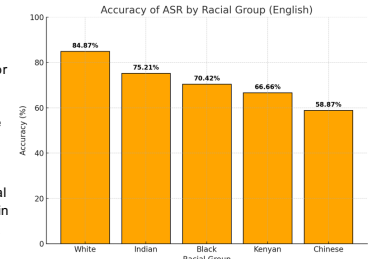
2. Understanding vs. Not Understanding Label



Result

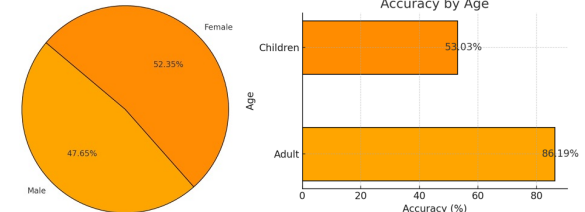
1. Racial Bias

This graph shows Alexa's speech recognition accuracy for the five different test races. Among them, the white group has the highest recognition accuracy reaching 84.87%. Our result clearly reveal issues with racial bias in Amazon Alexa system.



2. Gender & Age Bias

Distribution of ASR Accuracy by Gender



3. Biases in Different ASR Systems

